Discourse topicality and the processing of implicit causality in Chinese

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[**Introduction**] Pronoun resolution in causal events (e.g., *John disturbed Sam because **he**…*) can be affected by bottom-up information such as verb meaning. Verbs that implictly bias the subject (NP1) as the cause of a transitive event are called NP1-biased verbs, and verbs biasing the object (NP2) as the cause are called NP2-biased verbs. However, few studies have directly tested how top-down (discourse) cues influence pronoun resolution in causal events (but see [1], [2]; and [3] for implicit consequentiality) and often focused only on production or comprehension tasks. In this study, we explored how discourse topicality affects native Chinese speakers' pronoun resolution in causal events using both sentence completion and self-paced reading tasks.

To preview the findings, we discovered that discourse topicality effect was asymmetrical such that it only appeared in the production task but not during incremental sentence comprehension. We speculate that participants may rely more on bottom-up (verb) cues in resource-demanding real-time comprehension tasks than in offline production tasks where top-down (discourse) cues are relatively more accessible.

[**Methods**] We manipulated the factors VerbBias (NP1-biased/NP2-biased) and Topicality (Topic/Non-topic) across **Exp.1** (**sentence completion**, $N_{participant}$ = 38, $N_{target\text{-}item}$ = 24) and **Exp.2** (**self-paced reading**, $N_{participant}$ = 45, $N_{target\text{-}item}$ = 24). Eight verbs from [4] (4 NP1-biased, 4 NP2-biased) were used. Topicality was achieved through context (e.g., 'Let's talk about John.'). Only the subject (NP1) can be topical. In both experiments, the two referents are of different genders so that coreference with the subject (NP1 coreference) or the object (NP2 coreference) can be identified for Exp.1 and explicitly marked for Exp.2. Genders were counterbalanced across items. In Exp.2, the first two sentences were shown as a whole and only the last sentence was self-paced (followed by an antecedent choice task, accuracy rate by condition > 98%). See (1) for example target sets.

If discourse topicality modulates pronoun resolution, in Exp.1, discourse topicality is predicted to increase coreference with NP1 in NP1-biased conditions but lower coreference with NP2 in NP2-biased conditions. In Exp.2, reading times may be faster when the NP1 bias is strengthened by topicality but slower when NP2 bias conflicts with topicality (biasing NP1). Thus, we expect a VerbBias x Topicality interaction for both experiments.

[**Results**] Mixed effects logistic and linear regressions were used to analyze the coreference choices in Exp.1 and reading times in Exp.2, respectively. See **Fig.1-2** for results. In Exp.1, we found a VerbBias x Topicality interaction ($p < 0.001$) as expected: within the NP1-bias conditions, the probability of NP1 coreference was higher when NP1 was topical; but within the NP2-bias conditions, the probability of NP2 coreference was lower when NP1 – its 'competitor' – was topical. This topicality effect holds for almost all verbs, except for the NP1-biased verb 'shock.' In Exp.2, no effects reached significance at the critical pronoun and the following spillover region. Only at the final region was there a main effect of VerbBias ($p < 0.05$): sentences with NP2-biased verbs were processed faster. This NP2 advantage during incremental processing may reflect a recency effect right before the antecedent choice task when participants needed to recall the causer.

[**Discussion**] In addition to confirming the verb-related implicit causality effect in Chinese, the present study has found that discourse topicality influences causal coreference during (offline) language production. However, the discourse topicality effect was absent during incremental language comprehension. This may be because during the self-paced reading task, participants could not marshell sufficient cognitive resources for pronoun resolution and therefore prioritized bottom-up cues (i.e., verb semantics) over discourse-level cues (i.e., discourse topicality). Future work should explore whether the topicality effect emerges in less demanding tasks (visual world eye-tracking study in progress).

(1) **An example target item** for Exp.1 and Exp.2. Slashes indicate presentation regions in self-paced reading Exp.2. Sentences are transcribed to English (original sentences in Chinese).
Context: 'Let's talk about John$_{\textbf{topic}}$/yesterday$_{\textbf{non-topic}}$. John disturbed$_{\textbf{NP1-biased}}$/warned$_{\textbf{NP2-biased}}$ Mary.'
Critical sentence: 'This was/ because/ {he$_{NP1}$,she$_{NP2}$}/ in library/ loudly/ was reading./'

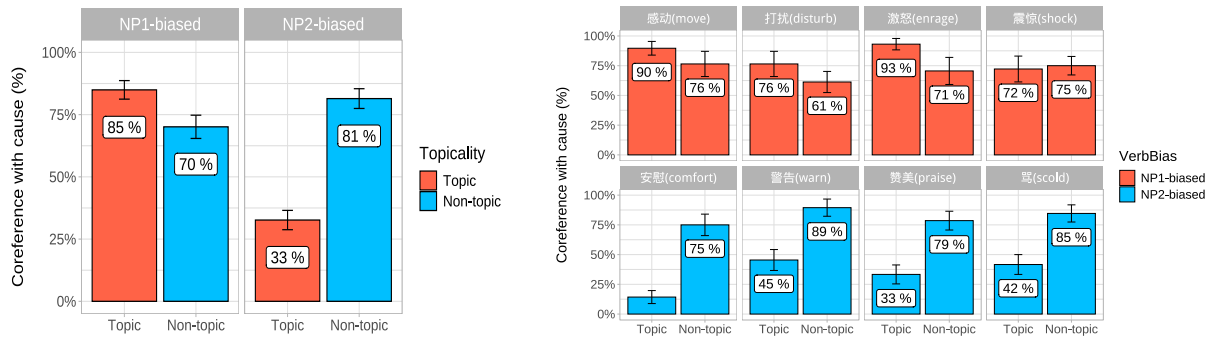

Fig.1: Mean coreference with cause referent collapsed over all verbs (left) and broken down by individual verbs (right) in Exp.1.

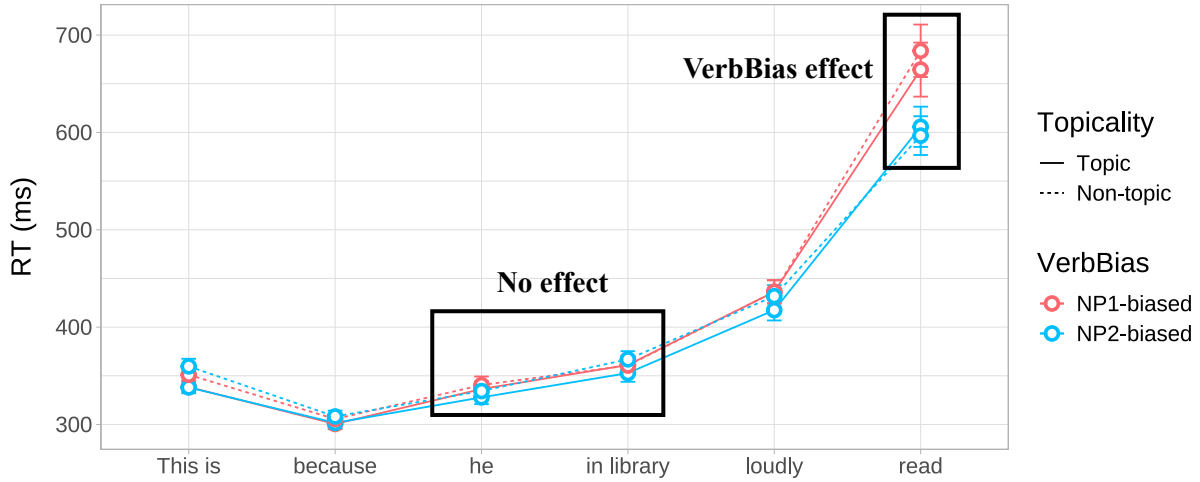

Fig.2: RTs by region and condition in the critical sentence in Exp.2.

References

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